

Apply apriori algorithm on given dataset, calculate support and confidence then find the association.

c1)

TID	Items
100	1 3 4
200	2 3 5
300	1 2 3 5
400	2 5

min support = 2.

C ₁		L ₁	
Itemset	Min. Supp.	Itemset	Min. Supp.
{1}	2	{1}	2
{2}	3	{2}	3
{3}	3	{3}	3
{4}	1	{5}	3
{5}	3		

C ₂		L ₂	
Itemset	Min. Supp.	Itemset	Min. Supp.
{1, 2}	1	{1, 3}	2
{1, 3}	2	{2, 3}	2
{1, 5}	1	{2, 5}	3
{2, 3}	2	{3, 5}	2
{2, 5}	3		
{3, 5}	2		

Itemset	Min. Supp.
{1, 2, 3}	1
{1, 3, 5}	1
{2, 3, 5}	2

Rule Generation :-

Association Rule	Support	Confidence	Confidence (C.I.)
$2 \wedge 3 \rightarrow 5$	2	$2/2 = 1$	100%
$3 \wedge 5 \rightarrow 2$	2	1	100%
$2 \wedge 5 \rightarrow 3$	2	$2/3 = 0.66$	66%
$2 \rightarrow 3 \wedge 5$	2	0.66	66%
$3 \rightarrow 2 \wedge 5$	2	0.66	66%
$5 \rightarrow 2 \wedge 3$	2	0.66	66%

(ii)

TID

Items

1

Bread, Milk

2

Bread, Diaper, Beer, Eggs

3

Milk, Diaper, Beer, Cola

4

Milk, Diaper, Beer, Cola

5

Bread, Milk, Diaper, Cola

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Here, Bread = 1, Milk = 2, Diaper = 3
 Beer = 4, Eggs = 5, Cola = 6

TID

Items

1

2

1 2

Items

Min. Supp.

2

1 3 4 5

S1y

3

3

1 3 4 6

S2y

4

4

1 3 4 6

S3y

4

5

1 3 4 6

S4y

3

1 2 3 6

S5y

1

S6y

3

C1

L1

Itemset

Min. Supp.

Itemset

Min. Supp.

S1y

3

S1 2y

2

S2y

4

S1 3y

2

S3y

4

S1 4y

1

S4y

3

S1 6y

1

S6y

3

S2 3y

3

S2 4y

2

S2 6y

3

S3 4y

3

S3 6y

3

S4 6y

2

itemset	Min. supp.	itemset	Min. supp.
$\{2, 3\}$	2	$\{2, 3, 4\}$	1
$\{2, 4\}$	2	$\{2, 4, 6\}$	0
$\{2, 6\}$	3	$\{2, 3, 6\}$	1
$\{3, 4\}$	2	$\{2, 3, 4, 6\}$	1
$\{3, 6\}$	2	$\{2, 4, 6\}$	1
$\{4, 6\}$	3	$\{2, 3, 4\}$	2
		$\{2, 3, 6\}$	3
		$\{2, 4, 6\}$	2
		$\{3, 4, 6\}$	2

L3

Scanned

→

itemset	Min. supp.	itemset	min. supp.
$\{2, 3, 4\}$	2	$\{2, 3, 4, 6\}$	2
$\{2, 3, 6\}$	3		
$\{2, 4, 6\}$	2		
$\{3, 4, 6\}$	2		

Scanned

C4

Association Rule	Confidence	Confidence (%)
$2 \rightarrow 3 \wedge 4 \wedge 6$	$2/4 = 0.5$	50%
$3 \rightarrow 2 \wedge 4 \wedge 6$	$2/4 = 0.5$	50%
$4 \rightarrow 2 \wedge 3 \wedge 6$	$2/3 = 0.66$	66%
$6 \rightarrow 2 \wedge 3 \wedge 4$	$2/3 = 0.66$	66%
$2 \wedge 3 \rightarrow 4 \wedge 6$	$2/3 = 0.66$	66%
$2 \wedge 4 \rightarrow 3 \wedge 6$	$2/2 = 0.1$	100%
$2 \wedge 6 \rightarrow 3 \wedge 4$	$2/3 = 0.66$	66%
$3 \wedge 4 \rightarrow 2 \wedge 6$	$2/3 = 0.66$	66%
$3 \wedge 6 \rightarrow 2 \wedge 4$	$2/3 = 0.66$	66%
$4 \wedge 6 \rightarrow 2 \wedge 3$	$2/2 = 0.1$	100%
$2 \wedge 3 \wedge 4 \rightarrow 6$	$2/2 = 0.1$	100%
$2 \wedge 4 \wedge 6 \rightarrow 3$	$2/2 = 0.1$	100%
$3 \wedge 4 \wedge 6 \rightarrow 2$	$2/2 = 0.1$	100%

Association Rule:

$2 \wedge 3 \wedge 6 \rightarrow 4$

Confidence

Confidence (1.)

$2/3 = 0.66$

66%

LAB
8

A.

Generate the Frequent Pattern tree for the given dataset.

TID	Items
1	E K M N O Y
2	D E K N O Y
3	A E K M
4	C K M U Y
5	C E I K D

Min. Support = 3

Step - 1

Item	Frequency
A	1
C	2
D	1
E	4
I	1
K	5
M	3
N	2
O	3
U	1
Y	3

Step - 2

TID	Sorted Items
1	K E M O Y
2	K E O Y
3	K E M
4	K M Y
5	K E O

$\{K:5, E:4, O:3, M:3, Y:3\}$

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min. supp.

1
0
1
1
1
2
3
2
2

min. supp

2

(e (90))

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Association Rule

21316 → 4

Confidence

Confidence (1.)

213 = 0.66

661.

A.

Generate the Frequent Pattern tree for the given dataset.

TID	Items	Min. Support = 3
1	E K M N O Y	
2	D E K N O Y	
3	A E K M	
4	C K M U Y	
5	C E I K D	

Step - 1

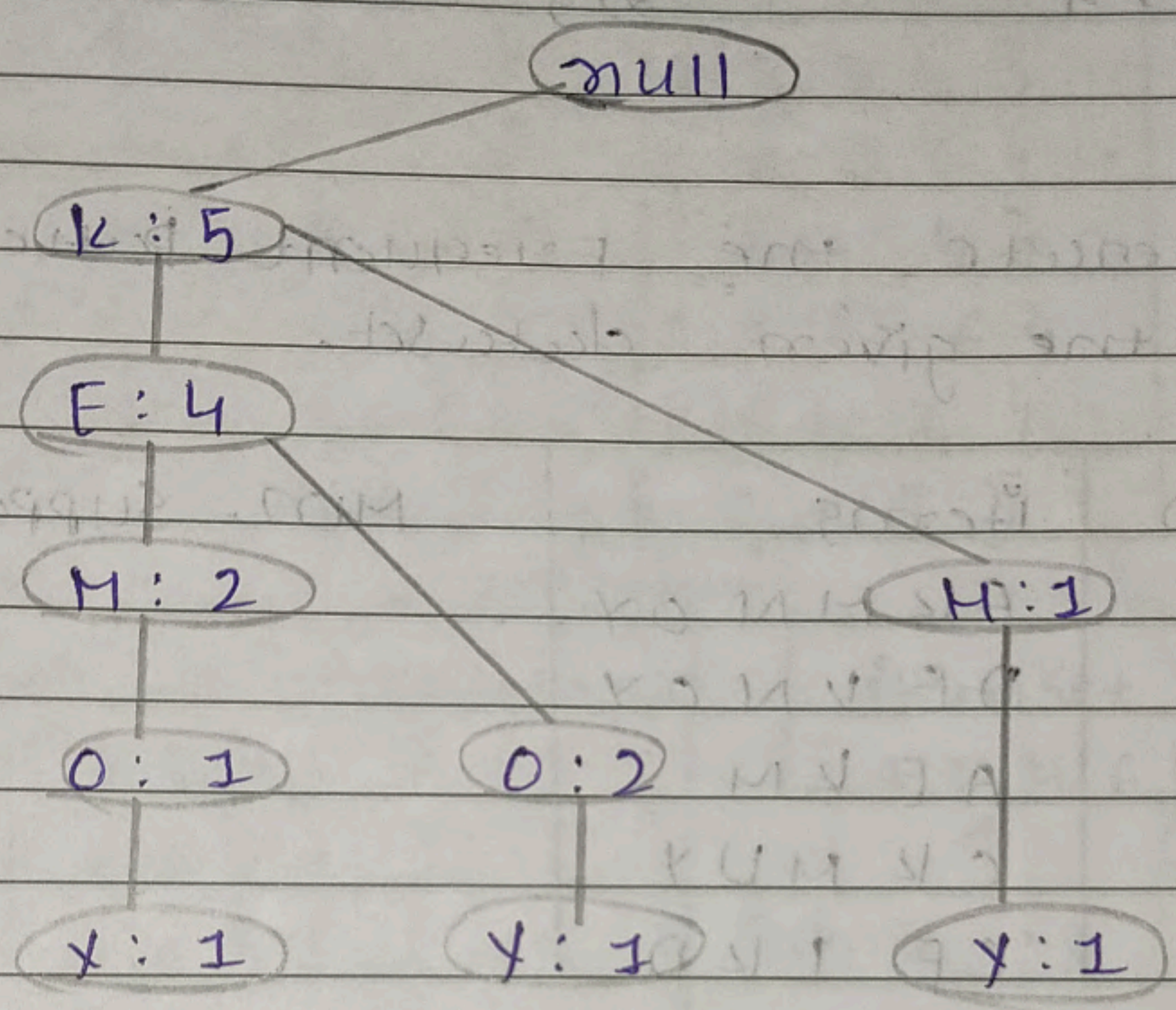
Item	Frequency
A	1
C	2
D	1
E	4
I	1
K	5
M	3
N	2
O	3
U	1
Y	3

Step - 2

TID	Sorted Items
1	K E M O Y
2	K E O Y
3	K E M
4	K M Y
5	K E O

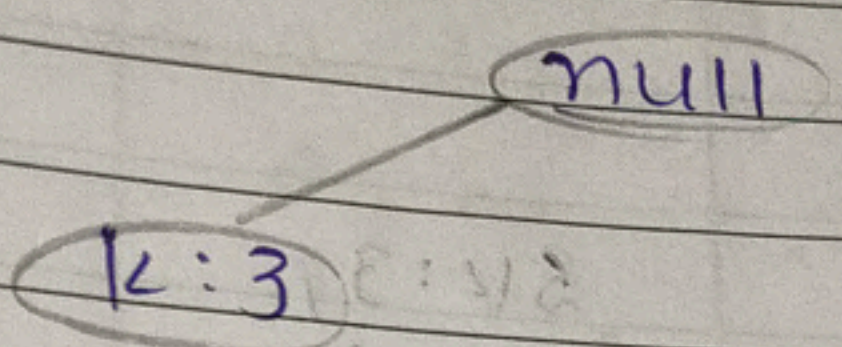
$K:5, E:4, O:3, M:3, Y:3$

Step-3 Building the F-P tree

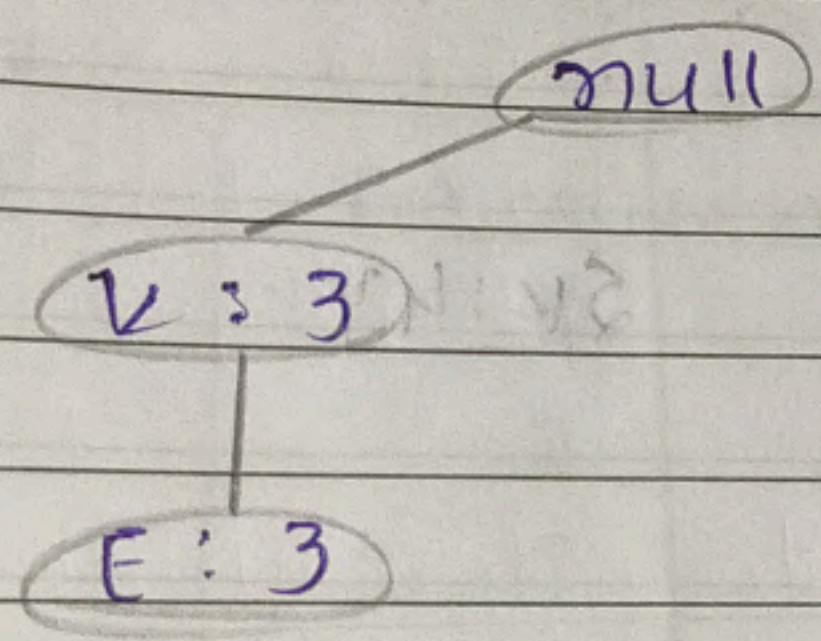


Item	Conditional Pattern Base
X	{KEMO:1} {KEO:1} {KM:1}
O	{KEM:1} {KE:2}
M	{KE:2} {K:1}
E	{K:4}
K	-

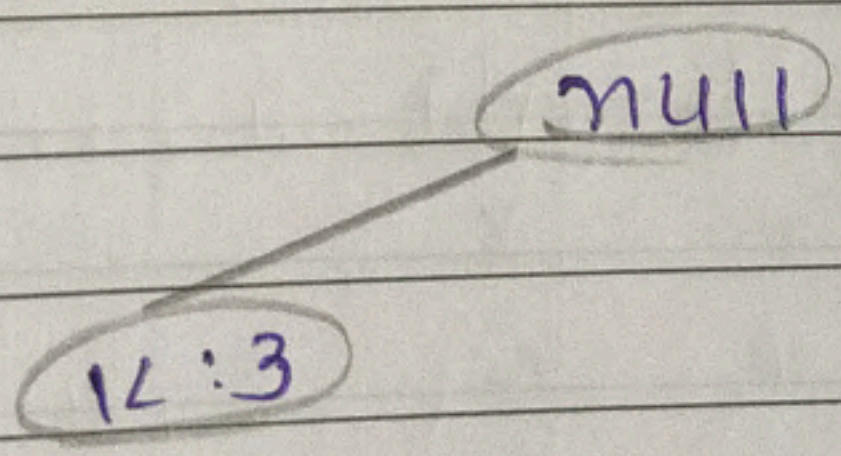
item	Conditional pattern Base	Conditional FP Tree
(i) x	$\$KEMO:1\$$ $\$KEO:1\$$ $\$KM:1\$$	$\$K:3\$$



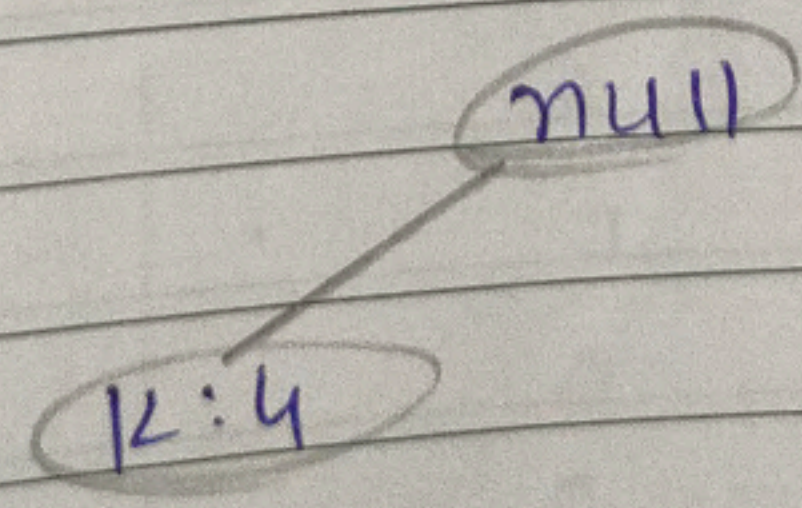
(ii) O	$\$KEM:1\$$ $\$KE:2\$$	$\$K:3\$$, $E:3\$$
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(iii) M	$\$KE:2\$$ $\$K:1\$$	$\$K:3\$$
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(iv) E	$\$K:4\$$	$\$K:4\$$
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Items	Conditional Pattern Base	Conditional FP tree	Frequent Pattern Generated
X	$\{KEMO:1\}$ $\{KEO:1\}$ $\{KM:1\}$	$\{K:3\}$	$\{K,4:3\}$
O	$\{KEM:1\}$ $\{KE:2\}$	$\{K:3, E:3\}$	$\{K, O:3\},$ $\{E, O:3\}, \{K, E, O:3\}$
M	$\{KE:2\}$ $\{K:1\}$	$\{K:3\}$	$\{K, M:3\}$
E	$\{K:4\}$	$\{K:4\}$	$\{K, E:4\}$
K	-	-	-

(ii)	TID	Items	Min Support = 2
1			
2		1 2 5	
3		2 4	
4		2 3	
5		1 2 4	
6		1 3	
7		2 3	
8		1 3	
9		1 2 3 5	
		1 2 3	

Step: 1

Item	Frequency
1	6
2	7
3	6
4	2
5	2

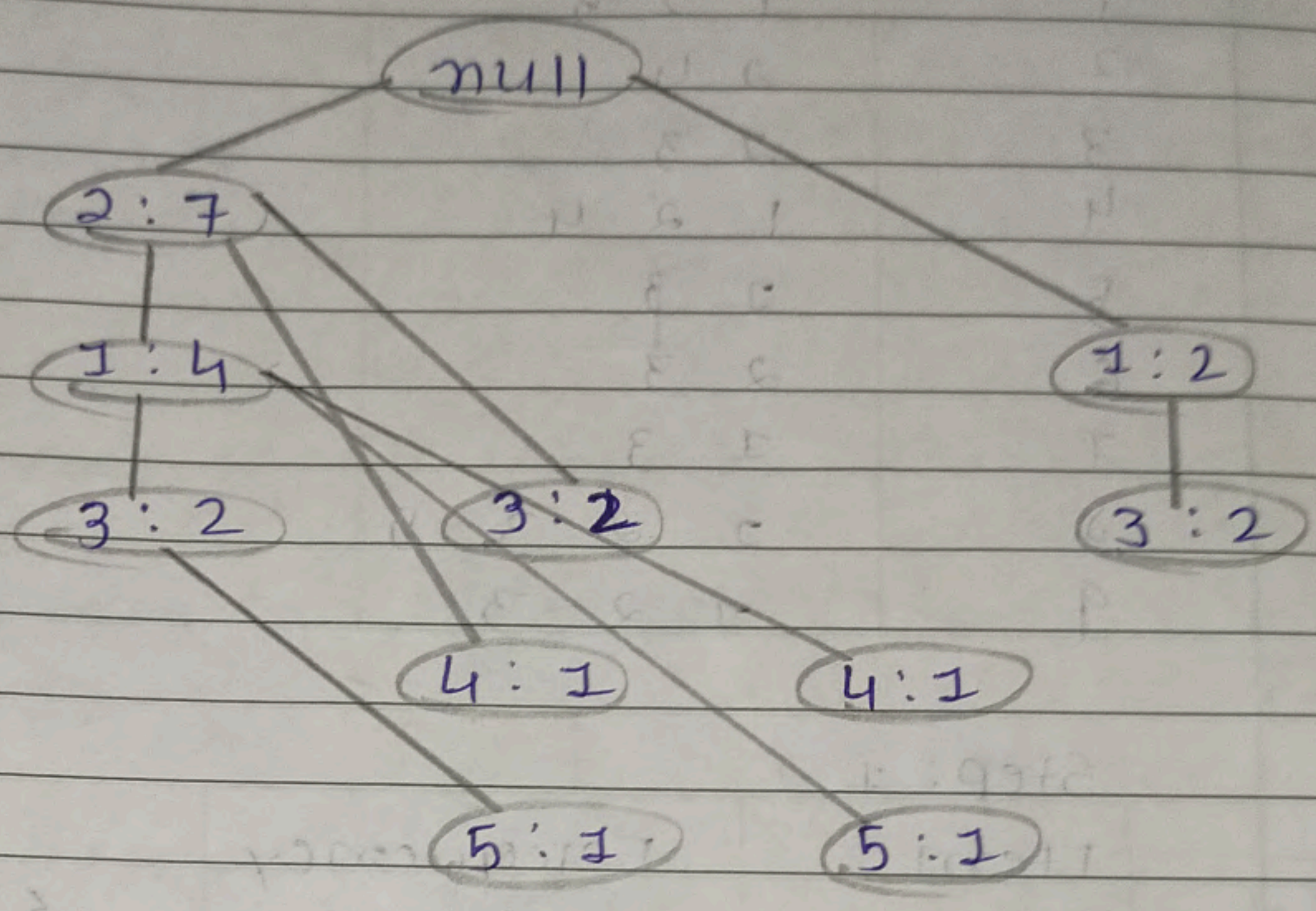
Step - 2

2:6, 2:7,
3:6, 4:2, 5:2

TID	Sorted Items
1	2 4 5
2	2 4
3	2 3
4	2 1 4
5	1 3
6	2 3
7	1 3
8	2 1 3 5
9	2 1 3

Step - 3

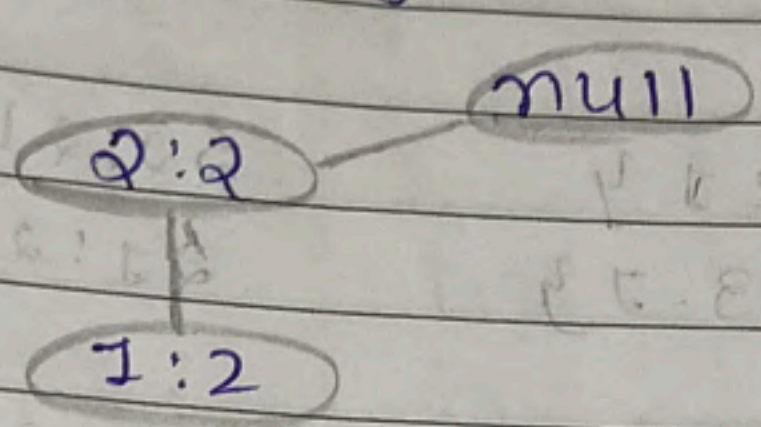
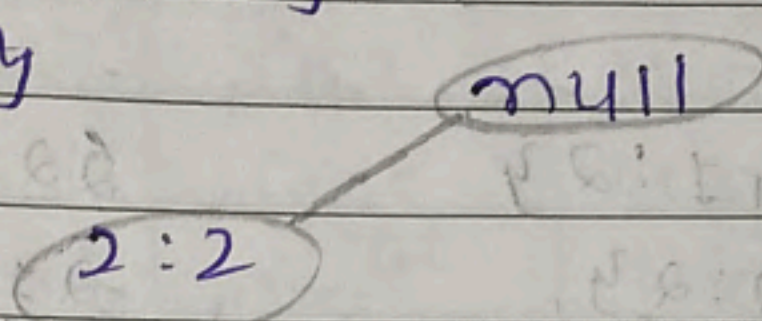
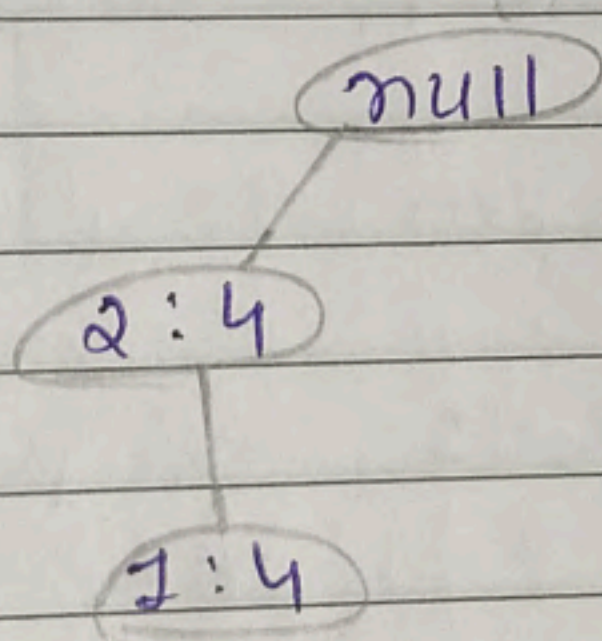
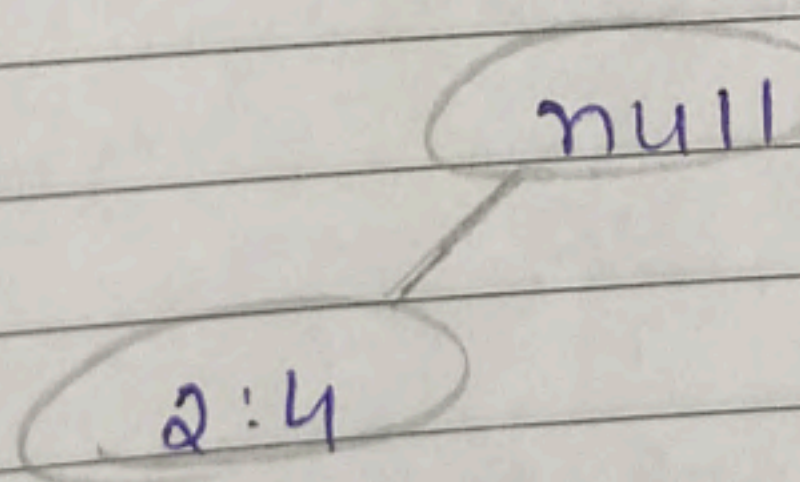
Building FP tree



Item	Conditional Pattern Base
5	$\{2, 1:1\}$ $\{2, 1, 3:1\}$
4	$\{2:1\}$ $\{2, 1:1\}$
3	$\{2, 1:1\}$ $\{2:1\}$ $\{1:1\}$
1	$\{2:4\}$
2	-

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	item	Conditional Pattern Base	Conditional EP
(i)	5	$\{2, 1: 1\}$ $\{2, 1\}$ $\{2, 1, 3: 1\}$ 	$\{2: 2, 1: 2\}$ $\{2: 2, 1: 2\}$
(ii)	4	$\{2: 1\}$ $\{2, 1: 1\}$ $\{2: 1\}$ 	$\{2: 2\}$ $\{2: 2\}$
(iii)	3	$\{2, 1: 2\}$ $\{2: 2\}$ $\{1: 2\}$ 	$\{2: 4, 1: 4\}$ $\{2: 4, 1: 4\}$
(iv)	4	$\{2: 4\}$ 	$\{2: 4\}$ $\{2: 4\}$

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Items	Conditional Pattern Base	Conditional FP tree	Frequent Pattern generated
5	$\{2,1\}:1y$ $\{2,1,3\}:2y$	$\{2\}:2y,$ $\{1\}:2y$	$\{2,5\}:2y, \{1,5\}:2y$ $\{2,1,5\}:2y$
4	$\{2\}:1y$ $\{2,1\}:1y$	$\{2\}:2y$	$\{2,4\}:2y$
3	$\{2,1\}:2y$ $\{2\}:2y,$ $\{1\}:2y$	$\{2\}:4y,$ $\{1\}:4y$	$\{2,3\}:4y, \{2,1\}:4y,$ $\{2,1,3\}:4y$
2	$\{2\}:4y$	$\{2\}:4y$	$\{2\}:4y$
1	-	-	-

B 9

A.

Apply Decision Tree for the given dataset.

Instance	a1	a2	a3	Classification
1	True	Hot	High	No
2	True	Hot	High	No
3	False	Hot	High	Yes
4	False	Cool	Normal	Yes
5	False	Cool	Normal	Yes
6	True	Cool	High	No
7	True	Cool Hot	High	No
8	True	Hot	Normal	Yes
9	False	Cool	Normal	Yes
10	False	Cool	High	Yes

→ For Entropy of all data

Distinct values in classification are Yes and No.

$$\text{Total :- } 6 + 4 = 10$$

$$\text{Entropy (D)} = \frac{-6}{10} \log_2 \left(\frac{6}{10} \right) - \frac{4}{10} \log_2 \left(\frac{4}{10} \right)$$

$$= 0.9709 \left[\because \log_y x = \frac{\log_{10} x}{\log_{10} y} \right]$$

→ Gain of a_1 :-

$$\text{Gain}(D, a_1) = \text{Entropy}(D) - \text{Entropy}(a_1)$$

Distinct values in a_1 True and False

	Yes	No	Total
True	1	4	5
False	5	0	5
			10

$$\text{Entropy}(a_1) = \frac{5}{10} \times \left[-\frac{1}{5} \log_2 \left(\frac{1}{5} \right) - \frac{4}{5} \log_2 \left(\frac{4}{5} \right) \right]$$

$$+ \frac{5}{10} \times \left[-\frac{5}{5} \log_2 \left(\frac{5}{5} \right) - \frac{0}{5} \log_2 \left(\frac{0}{5} \right) \right]$$

$$= 0.7219 \times \frac{5}{10} \quad \boxed{= 0.3609}$$

$$\text{Gain}(D, a_1) = 0.9709 - 0.3609$$

$$= 0.6099$$

=) Gain of a_2 :-

Distinct values in a_2	Yes	No	Total
Hot	2	3	5
Cool	4	1	5
			10

$$\text{Gain}(D, a_2) = 0.9709 -$$

$$\left\{ \frac{5}{10} \left[\frac{-2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right) \right] \right\}$$

Hot

$$+ \frac{5}{10} \left[\frac{-4}{5} \log_2 \left(\frac{4}{5} \right) - \frac{1}{5} \log_2 \left(\frac{1}{5} \right) \right]$$

Cool

$$= 0.1245$$

→ Gain for u_3 :

Distinct values in u_3	Yes	No	Total
High	2	4	6
Normal	4	0	4
			<u>10</u>

$$\text{Gain}(D, u_3) = 0.9709 -$$

$$\left\{ \frac{6}{10} \left[-\frac{2}{6} \log_2 \left(\frac{2}{6} \right) - \frac{4}{6} \log_2 \left(\frac{4}{6} \right) \right] + \right.$$

$$\left. \frac{4}{10} \left[-\frac{4}{4} \log_2 \left(\frac{4}{4} \right) \right] \right\}$$

$$= 0.4200$$

→ $\text{Gain}(D, u_1) = 0.6099$ → Maximum

$\text{Gain}(D, u_2) = 0.1245$

$\text{Gain}(D, u_3) = 0.4200$

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~

True (C1)

False

{1, 2, 6, 7, 8}

(Yes) {3, 4, 5, 9, 10}

New Data

Instance	C2	C3	Classification
1	Hot	High	No
2	Hot	High	No
6	Cool	High	No
7	Hot	High	No
8	Hot	Normal	No Yes

Entropy (D)

$$= -\frac{1}{5} \log_2 \left(\frac{1}{5} \right) - \frac{4}{5} \log_2 \left(\frac{4}{5} \right)$$

$$= 0.7219$$

Distinct value
in classification

Yes

No

Count

1

4

5

Gain of a_2

$$\text{Gain}(D, a_2) = \text{Entropy}(D) - \text{Entropy}(a_2)$$

Distinct values in a_2	Yes	No	Total
Hot	1	3	4
Cool	1	0	1
			5

$$\text{Gain}(D, a_2) = 0.7219$$

$$\left\{ -\frac{4}{5} \left[\frac{1}{4} \log_2 \left(\frac{1}{4} \right) - \frac{3}{4} \log_2 \left(\frac{3}{4} \right) \right] \right.$$

$$\left. + \frac{1}{5} \left[-\frac{1}{1} \log_2 \left(\frac{1}{1} \right) \right] \right\}$$

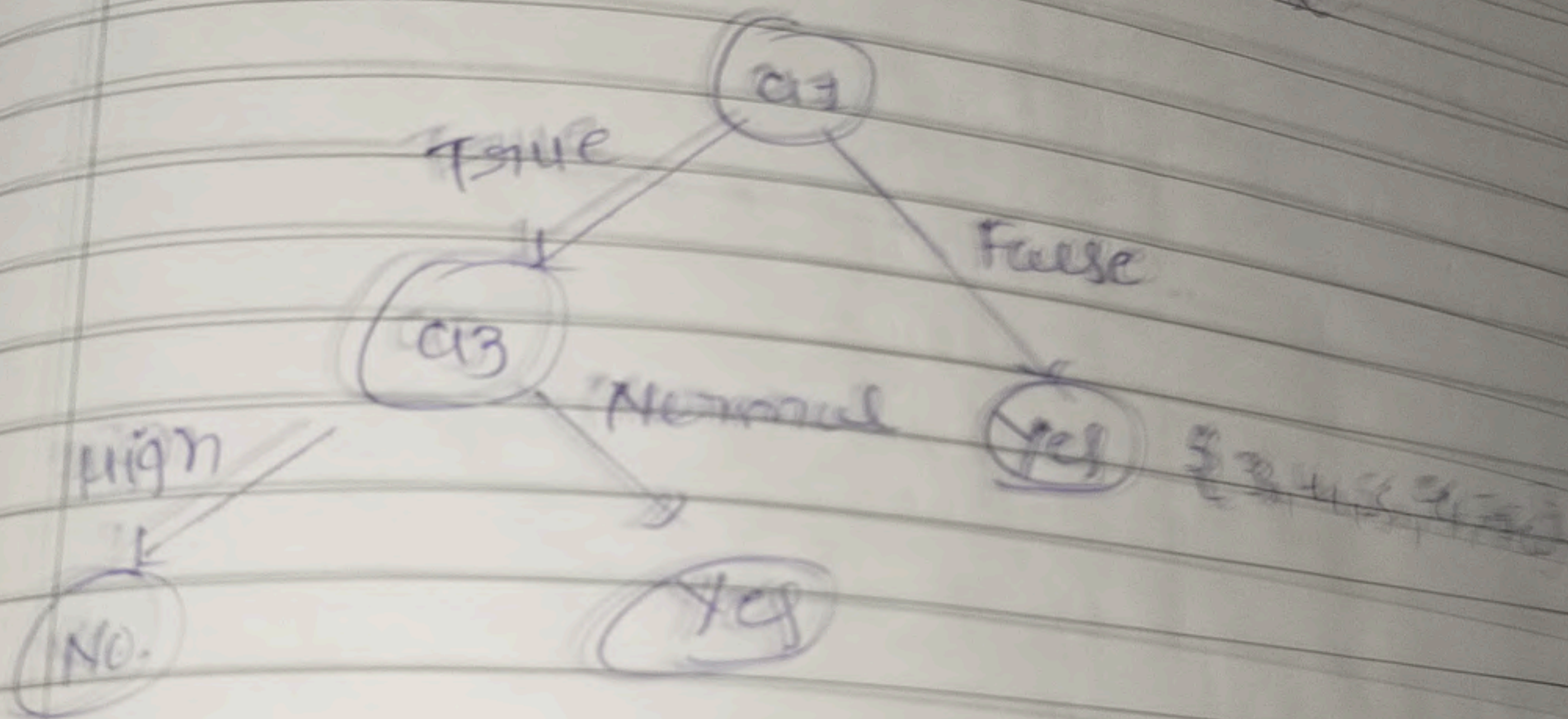
$$= 0.0729$$

Gain for a_3 :

Distinct values	Yes	No	Total
High	0	4	4
Normal	1	0	1
			5

Gain(D, a3) = 0.7219

$$\left\{ \frac{4}{5} \left[-\frac{4}{4} \log_2 \left(\frac{4}{4} \right) \right] + \frac{1}{5} \left[-\frac{1}{1} \log_2 \left(\frac{1}{1} \right) \right] \right\} = 0.7219$$



{1, 2, 6, 7}

{8}

LAB 11

Apply Naive Bayes classifiers for the given dataset.

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Day	Outlook	Temperature	Humidity	Wind	Play tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

$X' = (\text{outlook} = \text{Sunny}, \text{temperature} = \text{Cool}, \text{Humidity} = \text{High}, \text{wind} = \text{Strong})$ is it Yes or No? find it using naive Bayes classifier.

~> Solution :-

$$P(\text{Play} = \text{Yes}) = 9/14$$

$$P(\text{Play} = \text{No}) = 5/14$$

Outlook	Yes	No	Temp.	Yes	No
Sunny	2/9	3/5	Hot	2/9	2/5
Overcast	4/9	0/5	Mild	4/9	2/5
Rain	3/9	2/5	Cool	3/9	1/5

Humidity	Yes	No	Wind	Yes	No
High	3/9	4/5	Strong	3/9	3/5
Normal	6/9	1/5	Weak	6/9	2/5

$x' = (\text{Outlook} = \text{Sunny}, \text{Temperature} = \text{Cool}, \text{Humidity} = \text{High}, \text{Wind} = \text{Strong}) = 9$

$$P(c_i | x) = P(x_1 | c_i) \times P(x_2 | c_i) \times \dots \times P(x_n | c_i) \times P(c_i)$$

$$P(\text{Yes} | x') = [P(\text{Sunny} | \text{Yes}) * P(\text{Cool} | \text{Yes}) * P(\text{High} | \text{Yes}) * P(\text{Strong} | \text{Yes})] * P(\text{Play} = \text{Yes})$$

$$= [2/9 * 3/9 * 3/9 * 3/9] * 9/14 = 0.0053$$

$$P(\text{No} | x') = [P(\text{Sunny} | \text{No}) * P(\text{Cool} | \text{No}) * P(\text{High} | \text{No}) * P(\text{Strong} | \text{No})] * P(\text{Play} = \text{No})$$

$$= [3/5 * 1/5 * 4/5 * 3/5] * 5/14$$

$$= 0.0206$$

we label x' to be "No".

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B.

Apply

Gini Index for

the given dataset.

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RID	Age	Income	Student	Credit	Buyer Computer
1	Youth			excellent	no
2	Youth	high	no	fair	no
3	middle-age	high	no	excellent	no
4	Senior	high	no	fair	yes
5	Senior	medium	no	fair	yes
6	Senior	low	yes	fair	yes
7	middle-age	low	yes	excellent	no
8	Youth	low	yes	excellent	yes
9	Youth	medium	no	fair	no
10	Youth	low	yes	fair	yes
11	Senior	medium	yes	fair	yes
12	Youth	medium	yes	excellent	yes
13	middle-age	medium	no	excellent	yes
14	middle-age	high	yes	fair	yes
15	middle-age	medium	no	excellent	no
16	Senior	medium	no	excellent	no

buy-Computer

yes

9

5

NO

14

$$\text{Gini}(D) = 1 - \left(\frac{9}{14} \right)^2 - \left(\frac{5}{14} \right)^2 = 0.4592$$

Consider each of possible splitting subsets
for Age attribute

1. \$ youth, Middle y, \$ senior y
2. \$ youth, Senior y, \$ middle y
3. \$ middle, Senior y, \$ youth y

Distinct value	Yes	No	Total
youth	2	3	5
Middle	4	0	4
senior	3	2	5

Consider subset \$ youth, Middle y
Gini Age \$\in\$ \$ youth, Middle y (D)

$$= \frac{9}{14} \left(1 - \left(\frac{6}{9} \right)^2 - \left(\frac{3}{9} \right)^2 \right)$$

$$= \frac{5}{14} \left(1 - \left(\frac{3}{5} \right)^2 - \left(\frac{2}{5} \right)^2 \right)$$

$$= 0.4571$$

Gini Age \$\in\$ \$ youth, Senior y (D) =

$$\frac{10}{14} \left(1 - \left(\frac{5}{10} \right)^2 - \left(\frac{5}{10} \right)^2 \right) + \frac{4}{14} \left(1 - \left(\frac{4}{4} \right)^2 - \left(\frac{0}{4} \right)^2 \right)$$

$$= 0.3571$$

Gini Age \$\in\$ \$ Middle, Senior y (D) =

$$\frac{9}{14} \left(1 - \left(\frac{7}{9} \right)^2 - \left(\frac{2}{9} \right)^2 \right) + \frac{5}{14} \left(1 - \left(\frac{2}{5} \right)^2 - \left(\frac{3}{5} \right)^2 \right)$$

$$= 0.3937$$

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Similarly for income

	Yes	No	Total
{high, low} & {medium}			
{high, medium} & {low}			
{low, Medium} & {high}			
High	2	2	4
low	3	1	4
Medium	4	2	6

Gini Income $\in$ {high, low} (D)

$$= \frac{8}{24} \left(1 - \left(\frac{5}{8} \right)^2 - \left(\frac{3}{8} \right)^2 \right) + \frac{6}{24} \left(1 - \left(\frac{4}{6} \right)^2 - \left(\frac{2}{6} \right)^2 \right)$$

$$= 0.4583$$

Gini Income $\in$ {high, medium} (D)

$$= \frac{10}{24} \left(1 - \left(\frac{6}{10} \right)^2 - \left(\frac{4}{10} \right)^2 \right) + \frac{4}{24} \left(1 - \left(\frac{3}{4} \right)^2 - \left(\frac{1}{4} \right)^2 \right)$$

$$= 0.450$$

Gini Income $\in$ {low, Medium} (D)

$$= \frac{10}{24} \left(1 - \left(\frac{7}{10} \right)^2 - \left(\frac{3}{10} \right)^2 \right) + \frac{4}{24} \left(1 - \left(\frac{2}{4} \right)^2 - \left(\frac{2}{4} \right)^2 \right)$$

$$= 0.4429$$

For Student
Attribute,

	Yes	No	Total
yes	6	1	7
No	3	4	7

$$\begin{aligned} \text{Gini Student (D)} &= \frac{7}{14} \left(1 - \left(\frac{6}{7} \right)^2 - \left(\frac{1}{7} \right)^2 \right) \\ &+ \frac{7}{14} \left(1 - \left(\frac{3}{7} \right)^2 - \left(\frac{4}{7} \right)^2 \right) \\ &= 0.3675 \end{aligned}$$

For Credit
rating;

	Yes	No	Total
Fair	6	2	8
Excellent	3	3	6

Gini Credit rating =

$$\begin{aligned} &\frac{8}{14} \left(1 - \left(\frac{6}{8} \right)^2 - \left(\frac{2}{8} \right)^2 \right) + \frac{6}{14} \left(1 - \left(\frac{3}{6} \right)^2 - \left(\frac{3}{6} \right)^2 \right) \\ &= 0.4286 \end{aligned}$$

Attribute	Gini Index	Agini
Age	0.3571	$0.4592 - 0.3571 = 0.1021$
Income	0.4429	$0.4592 - 0.4429 = 0.0163$
Student	0.3673	$0.4592 - 0.3673 = 0.0919$
Credit rating	0.4286	$0.4592 - 0.4286 = 0.0306$

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Age
Youth Senior Middle aged

Income	Student	CR	Class
High	no	fair	no
High	no	ex.	no
Medium	no	fair	no
low	yes	fair	yes
Medium	yes	ex.	yes
Medium	no	fair	yes
low	yes	fair	yes
low	yes	ex	no
Medium	yes	fair	yes
Medium	no	ex	no

Income	Student	CR	Class
High	no	fair	yes
low	yes	excellent	yes
Medium	no	excellent	yes
High	yes	fair	yes

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Apply the k-means clustering on the following data and form the cluster for the value $k=2$.

Sr.	X	Y
1	1	1
2	1.5	2
3	3	4
4	5	7
5	3.5	5
6	4.5	5
7	3.5	4.5

Here, $k=2$ and we assume center is $(1,1)$ for cluster 1 and $(5,7)$ for cluster 2.

Data points		Distance to center		cluster
X	Y	1, 1	5, 7	
1	1	0	7.21	1
1.5	2	1.11	6.1	1
3	4	3.6	3.6	1
5	7	7.2	0	2
3.5	5	4.7	2.5	2
4.5	5	5.3	2.06	2
3.5	4.5	3	2.9	2

Euclidean Distance Formula :-

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

New Centroid :-

$$K_1 = (1.83, 2.33)$$

$$K_2 = (4.12, 5.375)$$

Data points		Distance to center		cluster	New cluster
X	Y				
1	1	1.83, 2.33	4.12, 5.375		
1.5	2	1.57	5.04	1	1
3	4	0.46	4.22	1	1
5	7	2.03	2.77	1	2
3.5	5	5.64	2.85	2	2
4.5	5	3.14	0.72	2	2
3.5	4.5	3.77	0.53	2	2
		2.74	1.07	2	2

New Centroid :-

$$K_1 = (1.25, 1.5)$$

$$K_2 = (3.9, 5.1)$$

Data points		Distance to center		cluster	New cluster
X	Y				
1	1	1.25, 1.5	3.9, 5.1		
1.5	2	0.55	5.02	1	1
3	4	0.56	3.92	1	1
5	7	3.05	2.42	2	2
3.5	5	6.66	2.19	2	2
4.5	5	4.15	0.141	2	2
3.5	4.5	4.77	0.26	2	2
		3.75	0.72	2	2

→ Here, we will stop as there is no change in the clusters.

Final Clusters :- 1 → {1, 2, 3}, 2 → {3, 4, 5, 6, 7}

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Perform

k - Medoids clustering

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S	X	Y
0	8	7
1	3	7
2	4	9
3	9	6
4	8	5
5	5	8
6	7	3
7	8	4
8	7	5
9	4	5

Step :- 1

let randomly selected 2 medoids, so
check $k=2$ and let $(1, 5)$ and
 $(2, 8)$ are the two medoids.

the dissimilarity of each non-medoid
point with medoids is calculated and
tabulated :-

S.N.	X	Y	Dissimilarity From c_1	Dissimilarity from c_2
0	8	7	$ 8-4 + 7-5 = 6$	$ 8-8 + 7-5 = 2$
1	3	7	3	7
2	4	9	4	8
3	9	6	6	2
4	5	8	4	6
5	7	3	5	3
6	8	4	5	1
7	7	5	3	1
8				

S.N.	X	Y	Dissimilarity y from c_1	Dissimilarity y from c_2
0	8	7	6	2
1	3	7	3	7
2	4	9	4	8
3	9	6	6	2
4	5	8	4	6
5	7	3	5	3
6	8	4	5	1
7	7	6	3	1

Each point is assigned to the cluster of that medoid whose dissimilarity is less.

the point 1, 2, 5 go to cluster c_1 and
0, 3, 6, 7, 8 go to cluster c_2 .

$$\text{the cost} = (3+4+4) + (2+2+3+1+1) = 20$$

Step:-03

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- randomly select one non-medoid point and recalculate the cost.
- let the randomly selected point be (8,4).
- the dissimilarity of each non-medoid point with the medoids - (4,5) and (2,8,4) is calculated and tabulated.

Sr.	X	Y	Dissimilarity y from c ₁	Dissimilarity y from c ₂
0	8	7	6	3
1	3	7	3	8
2	4	9	4	9
3	9	6	6	3
4	8	5	4	1
5	5	8	4	7
6	7	3	5	2
8	7	5	3	2

Each point is assigned to that cluster whose dissimilarity is less. So, the point 1, 2, 5 go to cluster c₁ and 0, 3, 4, 6, 8 go to cluster c₂.

$$\text{New Cost} = (3+4+4) + (3+3+1+2+2) = 22.$$

$$\begin{aligned} \text{Swap Cost} &= \text{New Cost} - \text{previous Cost} \\ &= 22 - 20 \\ &= 2. \end{aligned}$$

So, $2 > 0$ that is positive, now our previous medoid is best.

The total cost of Medoid $(8, 4) >$ the total cost when $(8, 5)$ was the medoid earlier & it generates the same cluster as earlier.